TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 8

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

Information:

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

Past paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Past Paper 8 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [MI/AI]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 35 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 5.817 – 2.75

Calculate: 8.717 - 5.65 = 3.067

Answer	3.067 ✓ [1 mark]

[1]

2. Work out 4 × 3.145

Round 12.583 to 2 decimal places = 12.58

Answer	12.58 ✓ [1 mark]

[1]

Past paper

Questions continue on the following page

Master Functional Skills maths - FSMUK Mathematics Level 2 Past Paper 8 Solutions © Training Qualifications UK 2023 All Rights Reserved

3. Convert 12 kilometres into miles.

Use 1.6 kilometres = 1 mile

Conversion: 1 mile ≈ 1.6 km --> 12 km ÷ 1.6 = 7.5 miles

Answer	7.5 ✓ miles	[2 marks]

[2]

4. There are 824 students at a school.

112 of them are in their final year.

One student is chosen at random.

What is the probability that this student is in their final year?

Give your answer as a fraction in its simplest form.

Fraction = 112 / 824. Divide numerator and denominator by 8:

112 ÷ 8 = 14 and 824 ÷ 8 = 103 --> 14 / 103

Answer	14 / 103 ✓	[2 marks]

[2]

Past paper

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5. Morgan is the manager of a concert arena.

The number of tickets sold in two years is shown:

Year 1:	**Year 2**:
784 693 tickets	523 154 tickets

Morgan thinks that the total number of tickets sold in these two years is more than 1 400 000

Is Morgan correct?

Show how you decide.

Total tickets sold = 784 693 + 523 154 = 1 307 847

Comparison: 1 307 847 < 1 400 000. Target was not reached. No.

Answer No (1 307 847 < 1. [2 marks]

[2]

Past paper

Questions continue on the following page

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6. Work out the original cost of this television:

Television
20% off
Now only £240

£240 represents (100% - 20%) = 80% of original price

Original price = £240 ÷ 0.80 = £300

Answer £ 300 ✓ *[2 marks]*

[2]

7. Work out:

$$8\frac{1}{4} + 6\frac{1}{5}$$

Give your answer as a mixed number.

Common denominator of 4 and 5 is 20:

8 1/4 + 6 1/5 = 8 5/20 + 6 4/20 = 14 9/20

Answer 14 9/20 ✓ *[2 marks]*

[2]

Past paper

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8. Dua is going to make a gold bracelet using 50 cubic centimetres (cm^3) of liquid gold.

Dua knows the density of gold is 19.3 g/cm^3.

How many grams of gold will Dua need to make the bracelet?

Use: $\text{Density} = \dfrac{\text{mass}}{\text{volume}}$

Formula: Mass = Density × Volume

Mass = 19.3 g/cm³ × 50 cm³ = 965 grams

Answer	965 ✓ grams [3 marks]

[3]

End of Section A.

Past paper

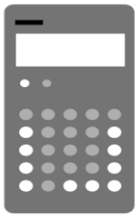
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Section B begins on Page 7.

Past paper

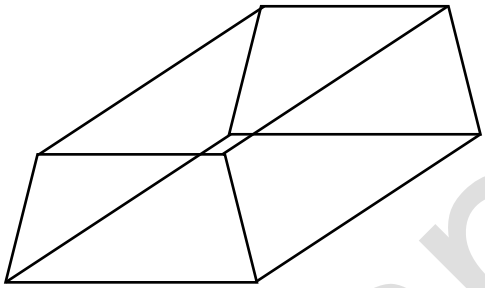
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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What is this shape?



a) Pyramid	**b)** Trapezium
c) Cuboid	**d)** Prism

Triangular prism ✓

Answer	[1 mark]

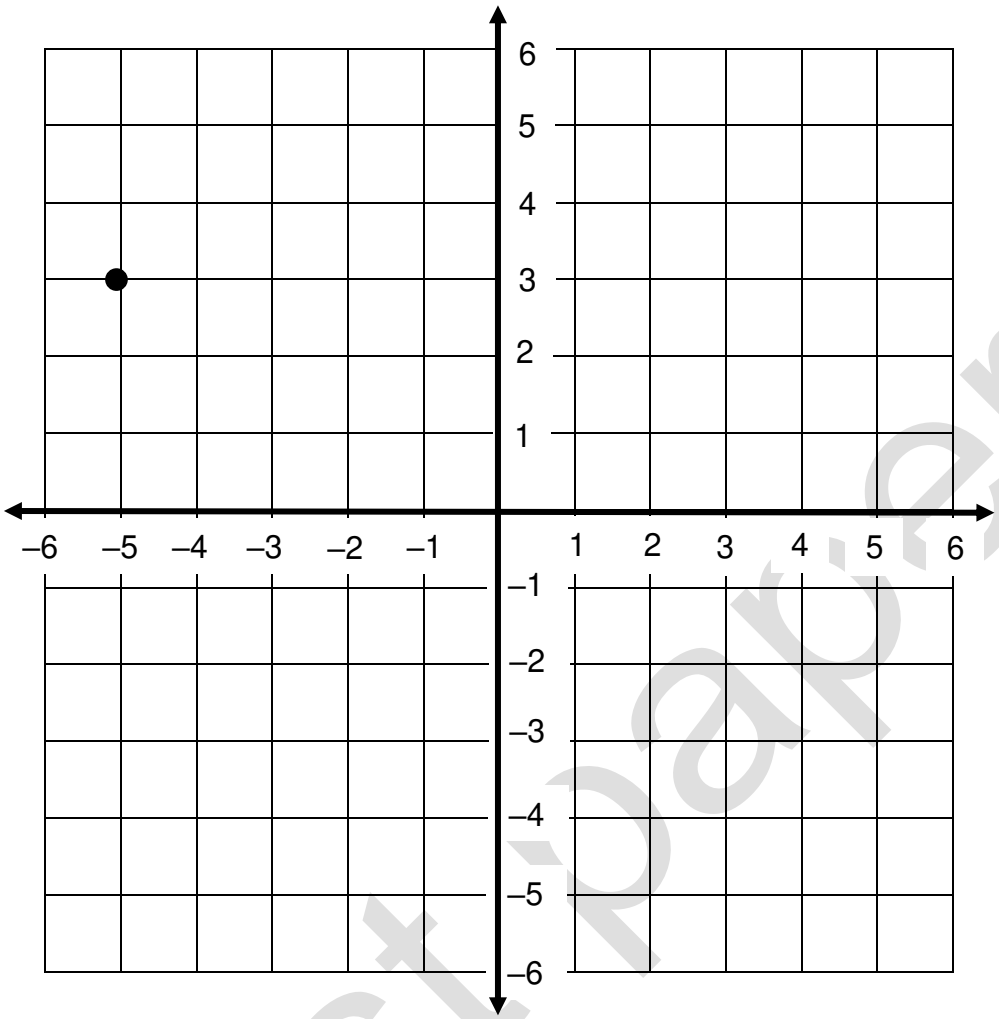
[1]

Past Paper

Questions continue on the following page

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2. Write down the coordinates of the point plotted on the grid.



Answer (−5, 3) (,) [1 mark]

[1]

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3. Zola wants a new sofa.

Zola can choose either a grey or blue sofa.

Both sofas have the same original price.

Grey sofa	Blue sofa
$\frac{1}{9}$ off	12% off

Which sofa has the bigger discount?

Show how you decide.

Blue proportion = 1/9 = 0.1111... ≈ 11.1%

Red proportion = 12% = 0.12 --> 11.1% < 12%. Blue had lower proportion.

Answer Blue (11.1% < 12%) ✓ [2 marks]

[2]

Past paper

Questions continue on the following page

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4. Work out:

$$\frac{4^2 + 40 \div 2}{15 - 3 \times 4}$$

Show your working.

BIDMAS: Powers and brackets: $4^2 \div 4 = 16 \div 4 = 4$ (or $15 - 3 = 12$)

Calculation: 12

Answer	12 ✓	[2 marks]

[2]

Past paper

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5. A local farm sells milk in glass bottles for £0.80 per pint.

The farm increases the price to £1.00 per pint.

Work out the percentage increase in price.

Percentage increase = ((1.0 - 0.8) ÷ 0.8) × 100 = 25%

Answer 25 ✓ [2 marks] %

[2]

6. Tai needs their house painted within 10 days.

It will take 3 painters 6 days to paint the house.

Will 2 painters working at the same rate get the house painted within 10 days?

Show how you decide.

Inverse proportion: 3 painters × 6 days = 18 person-days

2 painters take: 18 ÷ 2 = 9 days. Comparison: 9 ≤ 9 days. Yes.

Answer Yes (takes 9 days) ✓ [2 marks]

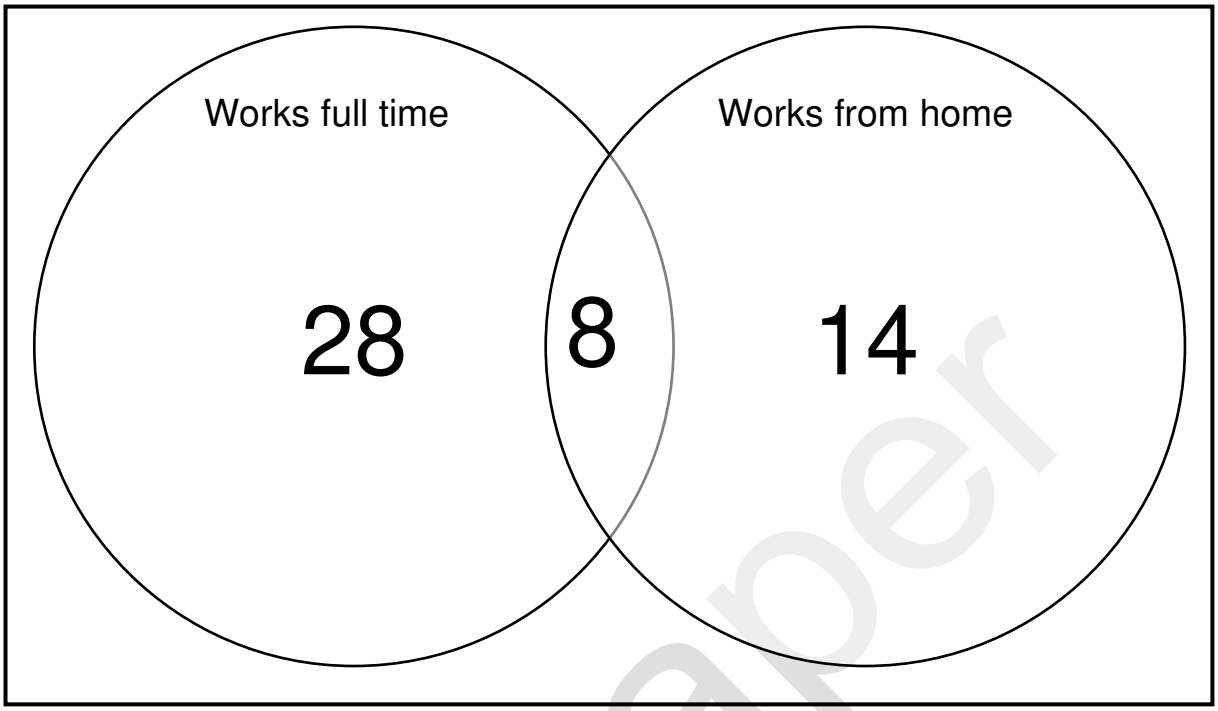
[2]

Past paper

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7. The Venn diagram shows the working patterns of a group of employees.



One of these employees is chosen at random.

What is the probability that this employee works full time?

Give your answer as a decimal **and** a percentage.

Probability = 36 / 50 = 72 / 100

Decimal = 0.72 | Percentage = 72%

Answer	a) Decimal: 0.72	b) Percentage: 72%	[3 marks]

[3]

Past paper

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8. Harper walks 15 kilometres in 200 minutes.

Harper thinks this is 4.2 kilometres per hour (kph).

Is Harper correct?

Show how you decide.

Speed = 15 km in 200 mins = (15 ÷ 200) × 60 = 4.5 kph

Comparison: 4.5 kph > 4.2 kph (walked faster than target). Therefore No.

Answer	No (4.5 kph > 4.2 ✓) [3 marks] kph

[3]

Past paper

Questions continue on the following page

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9. Charu needs to estimate the curved surface area of a hemisphere.

Charu knows:

- the radius of the hemisphere is 9 cm
- $\pi = 3.14159$

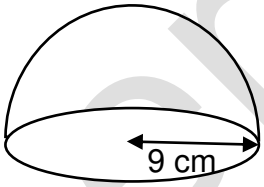
Charu rounds π to the nearest whole number to estimate the curved surface area.

What answer should Charu get?

Use:

$$A = \frac{4\pi r^2}{2}$$

Where:
A = curved surface area of sphere
r = radius of hemisphere



Not drawn to scale

Show your working.

Hemisphere Curved Surface Area = $4 \times \pi \times r^2 \div 2 = 2 \times \pi \times r^2$

Estimate using $\pi \approx 3$: $2 \times 3 \times 9^2 = 6 \times 81 = 486$ cm²

Answer	486 ✓	[3 marks] cm²

[3]

Past paper

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10. Drew needs to order 200 paper cups for a party.

Drew needs small, medium and large cups in the ratio 14 : 4 : 2

Drew thinks there will be 108 more small cups than large cups.

Is Drew correct?

Show how you decide.

Ratio Small : Medium : Large = 14 : 4 : 2 --> Total = 20 parts

1 part = 200 ÷ 20 = 10 --> Small = 140, Large = 20. Diff = 140 - 20 = 120

Comparison: 120 ≠ 108. Target was not met. Therefore No.

Answer No (Difference is ✓ [3 marks]

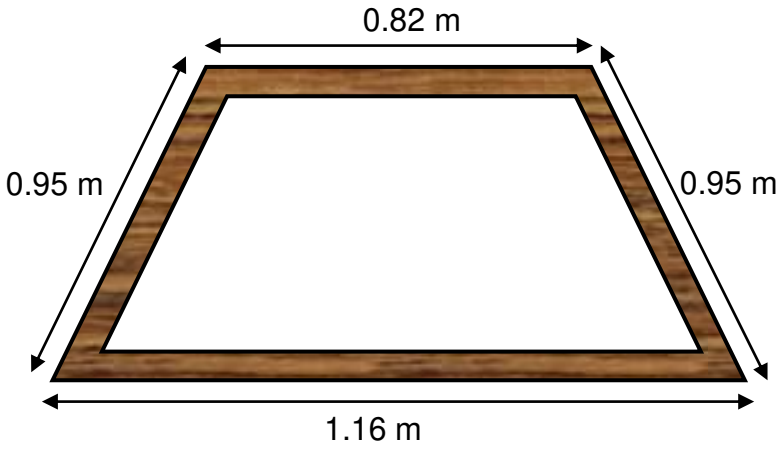
[3]

Past paper

Questions continue on the following page

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11. Kai wants to make a wooden picture frame with the dimensions shown below:



Not drawn to scale

Kai wants to buy 14% more wood than needed to allow for any mistakes.

Kai buys 4.5 metres of wood.

Has Kai bought enough wood?

Show your working.

Perimeter = 0.95 + 0.95 + 0.82 + 1.16 = 3.88 m

Increase by 14%: 3.88 × 1.14 = 4.4232 m

Comparison: 4.4232 m < 4.5 m. Roll is long enough. Yes.

Answer	Yes (4.42 m < 4.5 ✓) [4 marks]

[4]

Past paper

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12. Billie owns a café. Deepal is the café manager.

The table shows how many customers the café had over two weeks.

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	22	38	46	30	44
Week 2	23	40	45	42	20

Deepal thinks that on average there were more customers per day on week 1.

Billie disagrees.

Show how **both** could be correct.

Show your working.

Give reason here:

Sales Team A: Mean = 36, Median = 38

Sales Team B: Mean = 34, Median = 40

Comment: If using mean, Deepal is correct (36 > 34).

If using median, Billie is correct (40 > 38).

[4]

Past paper

Questions continue on the following page

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13. Ashley wants to buy a motorbike for £9750

Ashley paid £5000 into a new savings account two years ago.

The savings account paid 4% compound interest per year.

Ashley will use all the money in the savings account as a deposit towards the motorbike.

The rest will be paid in 40 equal monthly payments.

How much will each monthly payment be?

Deposit after 2 years: £5000 × $(1.04)^2$ = £5408

Remaining balance = £9750 – £5408 = £4342

Monthly payment = £4342 ÷ 40 months = £108.55

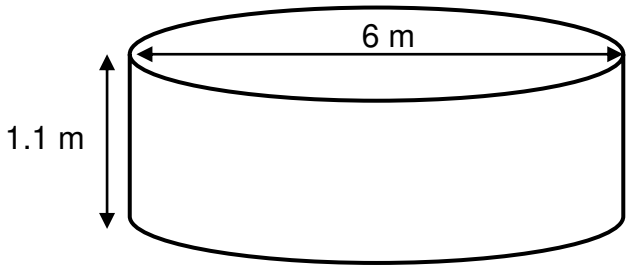
Answer	£ 108.55 ✓ [4 marks]

[4]

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14. Taylor has a fishpond in the shape of a cylinder.



Not drawn to scale

Taylor uses this formula to work out how many fish can be put into the pond.

$$\text{Maximum number of fish} = \frac{\text{volume (in m}^3\text{) of pond multiplied by 220}}{1000}$$

What is the maximum number of fish Taylor can put in the pond?

Use $\pi = 3.14$

Cylinder Volume = $\pi \times r^2 \times h = 3.14 \times 3^2 \times 1.1 = 31.086\ m^3$

Fish allowed = (31.086 × 220) ÷ 1000 = 6.838... fish --> Rounds down to 6 fish

Answer	6 ✓ [5 marks] fish

[5]

Questions continue on the following page

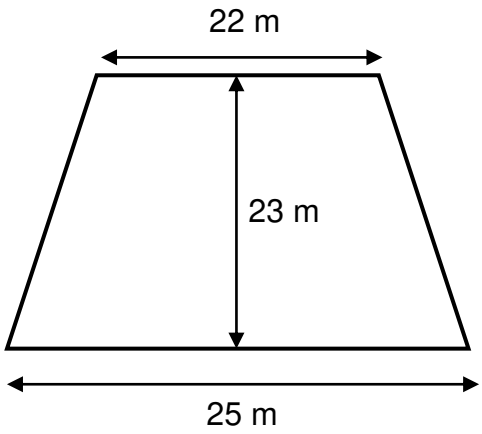
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15. Amory has turned a barn into an event venue.

Amory wants to estimate how much money the venue can be rented out for.

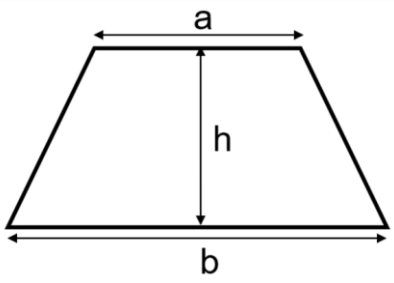
The floor plan of the venue is shown below:



Not drawn to scale

Use:

$$Area = \frac{(a+b)h}{2}$$



Amory estimates that one person will require 2.6 m^2 of floor space.

Amory finds these prices other local venues charge.

Venue	A	B	C	D	E
Price per person	£20	£35	£45	£30	£50

Trapezium Area = ((22 + 25) ÷ 2) × 23 = 23.5 × 23 = 540.5 m^2

People capacity = 540.5 ÷ 2.6 = 207.88... --> 207 people

Amory will charge the median price.

What is the maximum amount of money Amory can expect to make?

Show your working.

Answer box is on the following page

Past Paper

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Ordered ticket prices: 20, 25, 30, 35, 40, 45, 50 --> Median = £35

Total ticket revenue = 207 people × £35 = £7245

Answer	£ 7245 ✓	[6 marks]

[6]

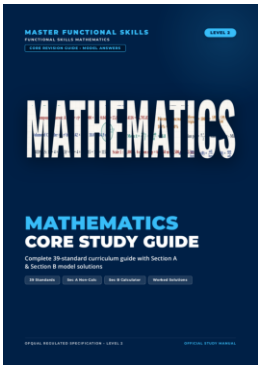
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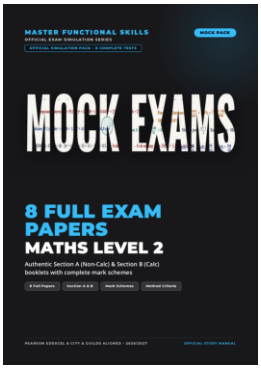


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